

Project 03 Initial Proposal

Introduction and topic and/or problem statement – A short introduction and summary of the goals of the project

We've all been there. Listening to the same song over and over again gets boring, and it's hard to find new, refreshing music. It is especially hard to find music that's similar in key, tempo, lyrics, and tone. That's why for this project, we want to create an application that recommends songs for you..introducing: StrumSense. A music recommender app perfect for all music lovers!

Our goal for this project is to chain multiple models together with public databases and APIs to create a model that recommends music.

Data sources that will be used – A reference to any datasets utilized in the project

Spotify, MusicBrainz, and other music databases/APIs may be utilized in this project. These will be useful for the model because it can check the key and tempo of a song, and recommend songs that have similar keys and tempos in the database. The music databases can also be used to fetch lyrics of the songs, which sentiment analysis can then be performed on them.

List of high-level methods, techniques and/or technologies that you are considering using.

For this project, we will start with a selected song and retrieve its metadata, including lyrics and musical features, from a music database such as MusicBrainz. The lyrics will be preprocessed and fed into an NLP model that performs sentiment analysis to determine the emotional tone of the song. This may include tags as "labels" to the song. Then, a semantic embedding model, such as a BERT-style encoder, will generate a vector representation of the lyrical meaning. These embeddings will then be compared against embeddings of other songs in the database using similarity search techniques like cosine or k-nearest neighbors. Based on this comparison, the system will return recommended songs that share similar sentiment, lyrical themes, key and bpm. Key and bpm will have been retrieved from the start based on the input song fetched from some type of music database.

Products to be delivered – what are the primary deliverables for the project? This is what we will be grading

The overall model will be one where you input a song title, and the output is a list of similar songs. The songs recommended should be similar based on similar sentiment, lyrical themes, key and bpm. The user should also be satisfied with the application and not encounter any errors.